

Extensive vs. Intensive Accumulation: Choice of Technique and Capacity Utilization

Preliminary CPR Estimation Results for the US Corporate Sector

Dissertation Chapter 2

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Bottom Line Up Front (BLUF)

- **Core Question:** How do distributive struggles and capital structure condition long-run productive capacity?
- **Methodology:** We resolve the $I(2)$ cointegration trap by treating the levels relationship as a **Cointegrating Polynomial Regression (CPR)** estimated via FM-OLS and IM-OLS.
- **Multicollinearity Solution:** Residual centering (orthogonalization) stabilizes the design matrix, reducing interaction VIFs from > 80.0 to exactly **1.0**.
- **Key Empirical Result:**
 - ▶ We find a **negative and highly significant** interaction between intensive accumulation (τ_t) and the wage share (ω_t) in the Fordist era, validating the induced innovation theory on a concave frontier.
 - ▶ A stark **structural break** occurs post-1974: the interaction becomes positive and statistically non-significant, reflecting the rise of financialization and global supply chains.

The Extensive Margin (K_t^{NRC})

Conceptual Definition

Nonresidential structures and infrastructure (K_t^{NRC}) represent the physical plant capacity, logistical networks, and spatial layout of production.

- **Temporal Properties:** High lumpiness, long gestation periods, and high irreversibility.
- **Asset Role:** In this framework, structures act as the direct physical scale parameter for capacity ($K_t^{scale} \equiv K_t^{NRC}$).
- **Hypothesized Sign:** $\theta > 0$. Expanding physical plants directly increases the maximum potential capacity of the system.

The Intensive Margin ($\tau_t \equiv K_t^{ME} / K_t^{NRC}$)

Conceptual Definition

The capital composition ratio (τ_t) represents the choice of technique—the machinery and equipment (K_t^{ME}) installed per unit of physical structure (K_t^{NRC}).

- **Temporal Properties:** Highly modular, shorter service lives, faster depreciation, and faster adaptation to technical change.
- **Asset Role:** Direct productive capacity capital that determines the technological intensity of the plant.
- **Hypothesized Sign:** $\psi > 0$. Higher machinery density increases the capacity payoff per unit of physical structure.

Technical Choice under Unbalanced Growth

The Technology Frontier

Productive capacity Y_t^P is generated on a concave technology frontier—not a neoclassical production function determining factor shares, but a physical capacity schedule mapping capital investment into maximum achievable output under distribution conflict:

$$Y_t^P = F(K_t^{NRC}, K_t^{ME}; \omega_t) = A_t (K_t^{NRC})^\theta (K_t^{ME})^{\psi + \lambda \omega_t}$$

- **Induced Innovation:** Wage pressure (ω_t) forces capitalists to increase machinery density ($\tau_t \equiv K_t^{ME} / K_t^{NRC}$) to save on labor costs.
- **Unbalanced Gestation:** The cost-minimization FOC yields:

$$\theta_t^\tau \equiv \frac{\partial \ln Y_t^P}{\partial \ln \tau_t} = \psi + \lambda \omega_t$$

- **Frontier Concavity:** Diminishing returns to mechanization imply:

$$\lambda \equiv \frac{\partial^2 \ln Y_t^P}{\partial \ln \tau_t \partial \omega_t} < 0$$

Econometric Specifications

Specification A: Single-Capital Aggregate (Shaikh-style)

$$\ln Y_t = \alpha + \beta_k \ln K_t^{cap} + \beta_\omega \omega_t + u_t$$

where $K_t^{cap} \equiv K_t^{ME} + K_t^{NRC}$ is the single capital aggregate.

Specification B: Composition-Mediated (Growth-Law Basis)

$$\ln Y_t = \alpha + \theta \tilde{k}_t^{NRC} + \psi \tilde{\tau}_t + \phi \tilde{\omega}_t + \lambda(\tilde{\tau}_t \cdot \tilde{\omega}_t)_{orth} + \epsilon_t$$

where variables are centered around their sample means.

- **Residual Centering (Orthogonalization):** To resolve severe multicollinearity, the interaction term is replaced by the residuals of its regression on the linear terms:

$$(\tilde{\tau}_t \cdot \tilde{\omega}_t)_{orth} = (\tilde{\tau}_t \cdot \tilde{\omega}_t) - \hat{\delta}_1 \tilde{k}_t^{NRC} - \hat{\delta}_2 \tilde{\tau}_t - \hat{\delta}_3 \tilde{\omega}_t$$

Cointegrating Polynomial Regression (CPR)

The $I(2)$ Cointegration Trap

Because $\tilde{\tau}_t$ and $\tilde{\omega}_t$ are both $I(1)$ processes, their product $\tilde{\tau}_t\tilde{\omega}_t$ contains $I(2)$ stochastic trends. Standard linear cointegration (e.g. Johansen) breaks down.

- **Asymptotic Corrections (FM-OLS & IM-OLS):**
 - ▶ **FM-OLS (Fully Modified OLS):** Adjusts for endogeneity and serial correlation in the residuals using a non-parametric long-run covariance matrix estimator.
 - ▶ **IM-OLS (Integrated Modified OLS):** Applies a parametric lead-and-lag correction to eliminate the endogenous feedback loop from the integrated regressors.
- **Trend Cancellation:** Cointegration is achieved because the $I(2)$ components in the interaction term cancel out with the non-linear dynamics of output, yielding an $I(0)$ stationary residual.
- **VIF Stabilization:** Residual centering drops the maximum VIF from over 80.0 in the raw interaction model to exactly **1.0**, stabilizing all parameters.

Results: Specification A (Single-Capital Baseline)

Table: Specification A: Single-Capital Aggregate (FM-OLS, Raw)

Window	Regressor	Coeff	t-stat	p-value
Golden Age (1945–1973)	Centered Aggregate stock (k_t^{cap})	0.9984	18.2751	0.0000
	Centered NFC wage share (ω_t)	-0.5824	-0.4052	0.6881
Post-War Fordist (1940–1973)	Centered Aggregate stock (k_t^{cap})	0.9821	24.1122	0.0000
	Centered NFC wage share (ω_t)	-0.7104	-0.5211	0.6045
Post-Fordist (1974–2024)	Centered Aggregate stock (k_t^{cap})	1.0543	35.8821	0.0000
	Centered NFC wage share (ω_t)	-4.8824	-2.8912	0.0051

- Scale coefficient is positive and robustly close to 1.0.
- **Robustness:** Cointegration confirmed; residuals reject the null of no cointegration under both Engle-Granger and Phillips-Ouliaris tests.

Results: Specification B (Composition-Mediated)

Window	Regressor	Coeff	t-stat	p-value
Golden Age (1945–1973)	Centered structures (k_t^{NRC})	0.2378	3.7248	0.0010
	Centered composition (τ_t)	0.3121	10.5266	0.0000
	Centered NFC wage share (ω_t)	-5.1605	-3.6241	0.0014
	Centered interaction, orth ($(\tau_t \cdot \omega_t)_{orth}$)	-1.0294	-2.4892	0.0201
Post-War Fordist (1940–1973)	Centered structures (k_t^{NRC})	0.0773	1.1122	0.2752
	Centered composition (τ_t)	0.3809	11.7582	0.0000
	Centered NFC wage share (ω_t)	-7.6639	-4.8891	0.0000
	Centered interaction, orth ($(\tau_t \cdot \omega_t)_{orth}$)	-2.0342	-4.9555	0.0000
Post-Fordist (1974–2024)	Centered structures (k_t^{NRC})	0.2146	4.2083	0.0001
	Centered composition (τ_t)	0.3520	16.9510	0.0000
	Centered NFC wage share (ω_t)	1.1260	0.7747	0.4425
	Centered interaction, orth ($(\tau_t \cdot \omega_t)_{orth}$)	0.0821	0.1257	0.9006

- Composition effect (τ_t) is highly significant. Interaction is negative and significant during the Fordist era.
- **Robustness:** Cointegration validated by both Engle-Granger and Phillips-Ouliaris tests, ensuring stationary residuals.

Critical Discussion: Validation vs. Structural Crisis

Validation of the Theory

The negative and highly significant interaction coefficient ($\lambda < 0$) in the Fordist era (1940 – 1973) strongly validates the induced innovation theory. Higher wage shares (ω_t) force capitalists to mechanize, but this drives the economy into the concave zone of lower marginal capacity conversion.

The Post-Fordist Structural Break

Post-1974, the interaction collapses to zero (0.0821, p-value 0.90).

- **Global Offshoring:** Domestic wage bargaining no longer drives technique choice in the same way because capital can relocate.
- **Financialization:** Capital accumulation shifts away from real plant investment to financial channels.
- **Off-shoring/Subcontracting:** The domestic wage share ω_t ceases to act as a reliable proxy for the total cost of labor involved in global production chains.

Unpacking Specification B: Golden Age Case Study (1945–1973)

Estimated Cointegrating Equation (FM-OLS)

$$y_t = 14.8617 + 0.2378\tilde{k}_t^{NRC} + 0.3121\tilde{\tau}_t - 5.1605\tilde{\omega}_t - 1.0294(\tilde{\tau}_t \cdot \tilde{\omega}_t)_{orth}$$

- **State-Dependent Capacity Payoff of Composition (θ_t^τ):**

$$\theta_t^\tau \equiv \frac{\partial y_t}{\partial \tilde{\tau}_t} = 0.3121 - 1.0294\tilde{\omega}_t$$

- **Numerical Sensitivities of Wage Share Variations:**

- ▶ **Baseline (Average ω_t):** $\tilde{\omega}_t = 0 \implies \theta_t^\tau = \mathbf{0.3121}$.
- ▶ **High Wage Share ($\tilde{\omega}_t = +0.02$):** $\theta_t^\tau = 0.3121 - 1.0294(0.02) = \mathbf{0.2915}$ (payoff falls by $\approx 6.6\%$).
- ▶ **Low Wage Share ($\tilde{\omega}_t = -0.02$):** $\theta_t^\tau = 0.3121 - 1.0294(-0.02) = \mathbf{0.3327}$ (payoff rises by $\approx 6.6\%$).

- **Economic Intuition:** Higher distributive wage pressure forces capitalists to accelerate mechanization, but drives the system into a concave region of the frontier, lowering the marginal capacity payoff of machinery.

Capacity Utilization & Elasticity Reconstruction (1945–1973)

- **Sectoral Division:** Elasticities estimated on NFC data; utilization (μ_t) reconstructed relative to **Total Real GDP**.
- **Historical Path Comparison:**
 - ▶ **Spec B:** Captures utilization fluctuations ($\sigma_\mu = 12.98\%$), anchored at 1973 peak ($\mu_{1973} = 1.0$).
 - ▶ **Shaikh-Style:** Higher volatility ($\sigma_\mu = 15.06\%$) due to the constant scale assumption.
 - ▶ **HP Filter:** Dampens real capacity squeezes ($\sigma_\mu = 3.79\%$) by absorbing demand booms into trend.
- **Cross-Method Correlation:** Spec B correlates with Shaikh ($\rho = 0.70$) and HP cycle ($\rho = 0.40$).
- **Mechanization Bias & Cushioning:**
 - ▶ Structures elasticity is negative ($\theta_t^{NRC} \approx -0.07$): empty plants without machinery yield negative capacity payoff.
 - ▶ Aggregate elasticity (θ_t^{total}) is positive and dynamic (**0.84–2.30**).
 - ▶ In the 1960 slowdown, Spec B ($\mu_{1960} = 1.19$) cushions the Shaikh collapse ($\mu_{1960} = 0.86$) via slow capacity envelope expansion.

Conclusion & Next Steps

- We have established a robust, cointegrated long-run relationship using Cointegrating Polynomial Regression (CPR) and residual centering.
- The results confirm that capital composition (τ_t) mediates capacity conversion, and that this mediation is conditioned by distribution (ω_t) in the Fordist core.
- **Reconstructing Capacity & Utilization:**
 - ▶ Reconstruct the potential capacity output series:

$$\hat{y}_t^P = \hat{\alpha} + \hat{\theta} \tilde{k}_t^{NRC} + \hat{\psi} \tilde{\tau}_t + \hat{\phi} \tilde{\omega}_t + \hat{\lambda} (\tilde{\tau}_t \tilde{\omega}_t)_{orth}$$

- ▶ Recover the latent capacity utilization series as the stationary cointegrating residual:

$$\widehat{\ln \mu_t} = y_t - \hat{y}_t^P$$